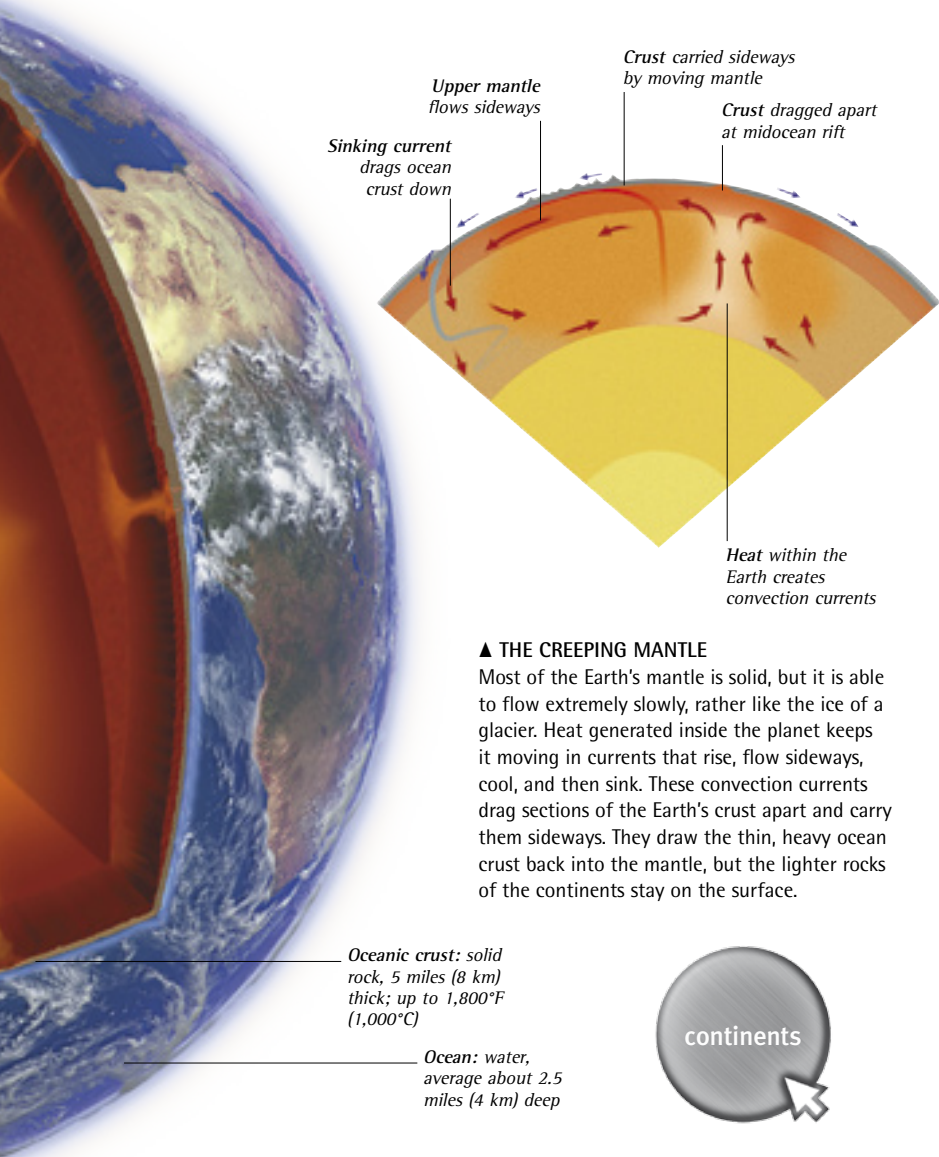




▲ THE CREEPING MANTLE

Most of the Earth's mantle is solid, but it is able to flow extremely slowly, rather like the ice of a glacier. Heat generated inside the planet keeps it moving in currents that rise, flow sideways, cool, and then sink. These convection currents drag sections of the Earth's crust apart and carry them sideways. They draw the thin, heavy ocean crust back into the mantle, but the lighter rocks of the continents stay on the surface.

Oceanic crust: solid rock, 5 miles (8 km) thick; up to 1,800°F (1,000°C)

Ocean: water, average about 2.5 miles (4 km) deep

continents

CONTINENTAL DRIFT

As the Earth's crust is dragged around by the mobile mantle, the slabs of rock that form the continents are dragged around with it. This makes the continents and oceans change in size and shape.



PANGAEA

Some 200 million years ago there was just one huge supercontinent that geologists call Pangaea. This was surrounded by a global ocean known as Panthalassa.



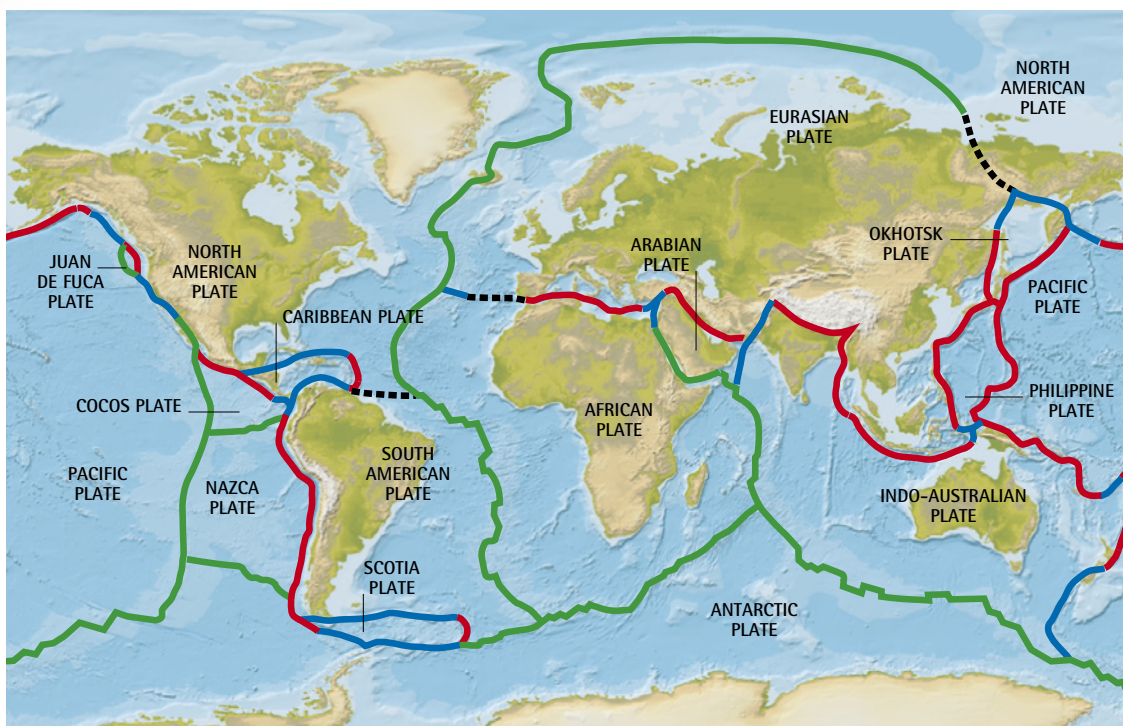
BREAKUP

By 100 million years ago, America and Africa had drifted apart, opening up the Atlantic Ocean; India was moving north toward Asia.



TODAY'S OCEANS

During the next 100 million years the Atlantic became broader as America moved west. Panthalassa shrank to become the Pacific, while India drifted north to collide with Asia.



◀ MOVING PLATES

The creeping mantle effect has made the Earth's crust break into many separate plates that are constantly on the move. They are either slipping apart at the divergent boundaries that form midocean ridges, pushing together at the convergent boundaries that are marked by the deep ocean trenches, or sliding past each other at transform boundaries. These plate boundaries are the sites of frequent earthquakes and tsunamis, and they are dotted with volcanoes and volcanic islands.

KEY

- Convergent boundary
- Divergent boundary
- Transform boundary
- Uncertain boundary